15 Intricate palimpsests. Disentangling Italian cultural landscapes through open and drone-borne LiDAR

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A recent systematic review of twenty years of LiDAR in archaeology shows that results in Mediterranean contexts remain comparatively limited (Vinci et al., 2025). Two factors play a key role. First, long and intensive occupation has turned many landscapes into complex palimpsests, in which anthropogenic and natural features are hard to disentangle. Second, coverage of open, freely accessible LiDAR data remains patchy, hindering landscape analysis of entire regions of dense past settlement, such as the Aegean.

This paper presents case studies from Italian contexts, mainly NE Italy, across diverse geomorphological settings from alluvial and coastal plains to Karst, where LiDAR has been extensively applied to study past human-environmental interactions. Raw data and DTMs were processed with FLOSS tools (QGIS, CloudCompare, R, Relief Visualization Toolbox), ensuring transparent, reproducible workflows and enabling public use and outreach.
Using equipment recently acquired by the Department of Geosciences (University of Padova), ad hoc drone-borne acquisitions were carried out in recent years in other regions where high-resolution open LiDAR data were not available. Where possible, we compared them with open regional datasets. The comparison reveals a clear trade-off. Drone surveys offer greater granularity, but are costly, time-consuming and realistically cover only limited portions of territory. Open datasets, though generally coarser, cover extensive areas and allow a shift from site-based to landscape-scale analysis. Moreover, where raw point clouds are released, they can be reprocessed into ad hoc Digital Feature Models (sensu Štular et al., 2020) tailored to specific evidence and areas.

Overall, the results support a multi-proxy, multi-temporal and multi-scale approach integrating LiDAR with historical cartography and geoarchaeological fieldwork, which in several cases was essential to understand land-use dynamics and long-term landscape transformations.